

SUBTRACT, TRANSPORT, OR REPLAY? AUDITABLE DELETION FROM LANGUAGE-MODEL MEMORY

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ABSTRACT

Exact deletion from persistent language-model memory depends on whether a record’s effect remains addressable after later computation. Native Kimi Delta Attention (KDA) gives a negative result for the tested receipt interface: the corpus-pooled raw recurrent contribution changes by 12–49% with the suffix and remains 8–49% after a decay-ledger correction. Native omission also changes later transition and write terms and other active caches. Frozen-input transport succeeds on its fixed-input control; the changed terms place native omission outside the tested receipt classes. Checkpoint replay supplies the evaluated recomputation path; zero residual on final logits and all 80 audited KDA arrays verifies restoration across the declared checkpoint surface.

The complementary result is constructive. We retrofit support-vector memory into frozen Gemma 3 without attention transfer, low-rank recovery, distillation, adapters, or language-model parameter updates. Prefix-mass preservation and one box per prefix solve give base-matched admission at 4B with 1.85% perplexity overhead. At 1B and 4B, verified deletion agrees with its conditional retained-key refit within 1.3×10^{-10} maximum next-token KL; behavioral attacks at 4B reach never-stored or chance baselines. Across 1B, 4B, and 12B, the 4B checkpoint uniquely combines base-matched admission with low overhead. The paper’s two contributions are a negative result for native KDA’s tested receipt classes and a positive training-free construction for addressable pretrained memory.

1 INTRODUCTION

Imagine that an assistant has already compressed yesterday’s conversation into a long-term state. The original text is no longer being re-read, so deleting a sentence from a visible prompt does not delete what the assistant already stored. The stakes are concrete in clinical documentation. An ambient scribe hears “my mother had breast cancer” during history taking; the statement enters the assistant’s persistent memory and begins shaping everything downstream—the family-history section, the risk assessment, the screening plan. When the patient later clarifies that imaging found a benign lump, appending the correction is not enough: both statements now compete inside the memory, and the superseded one can resurface in a summary written weeks later. A useful memory deletion must instead edit the state itself and answer a counterfactual question: *does the edited memory match the memory we would have built had this record never been included?* This paper asks which memory representations make that answer attainable without always rebuilding from the beginning.

Most unlearning work edits model weights and measures whether a target stops being recalled (TOFU, MUSE, and WMDP (Maini et al., 2024; Shi et al., 2025; Li et al., 2024a)). Such suppression can be reversed or exposed by representations, relearning, sampling, or pre/post snapshots (Xu et al., 2026; Hu et al., 2024; Thaker et al., 2024; Lynch et al., 2024; Wu et al., 2025). We instead study unlearning by design: after an autoregressive model has ingested a record into persistent state, what operation makes that instantiated memory match a declared record-omitted reference? MUNKEY applies the same structural premise to an external image-classification bank (Laguna et al., 2026); our substrate is in-context language-model memory.

An edit is *auditable* when its state or output can be checked against a declared record-omitted reference at a stated numerical tolerance. We ask two narrower questions. Can native recurrent state be deleted by a receipt fixed at the record boundary or transported through later updates? And can addressable memory be installed in a pretrained model without conversion training?

Native Kimi Delta Attention (KDA) supplies the negative result. Across three corpora, a record’s recurrent contribution varies by 12–49% with the suffix. Full affine transport reproduces a deliberately frozen suffix to 6.1×10^{-6} maximum relative error. Native omission changes later transition and write terms: in six source-free configurations, the median forcing-to-difference norm ratio is 0.84, and convolution and MLA caches also change. A receipt saved at the record boundary therefore cannot be carried along the record-present suffix to reproduce the omitted state. That places native omission outside the state-and-receipt interface we tested. Checkpoint replay recomputes the surviving suffix; zero residual on final logits and all 80 audited KDA arrays verifies restoration across the declared checkpoint surface.

Frozen Gemma supplies the positive result. We retrofit the companion support-vector gate into Gemma 3’s frozen global layers (Ramesh, 2026). Prior hybrid conversions train around a replacement (Zhang et al., 2024; Wang et al., 2024); this retrofit holds all language-model parameters fixed. Prefix-mass preservation and a separate $C_b = 1/(\nu n_b)$ for each prefix solve make the pretrained read-out usable without attention transfer, distillation, or recovery training. On the untouched confirmation cohort, the 4B graft admits the same 6/8 records as the base model with 1.85% perplexity overhead. At 1B and 4B, verified deletion and retained-key refit agree within 1.3×10^{-10} maximum next-token KL. Among the corrected 1B graft, the 4B graft, and the predeclared 12B second checkpoint, 4B alone combines base-matched admission with low overhead; behavioral conclusions therefore use 4B.

Scope. Our deletion target is persistent in-context state; pretrained weights and previously observed outputs remain unchanged (Pawelczyk et al., 2024; Wu et al., 2025). Gemma’s certificate compares decrement with a refit over the same contextualized keys. Reaching the imprint-free reference uses full repack—re-ingesting the edited raw context from the beginning—to remove ingestion-time imprint on neighboring keys. Kimi replay rebuilds its native logical recurrent and attention state. Each reference certifies a distinct target.

Our contributions are:

1. **A negative result for native KDA receipts.** Native omission lies outside the static and decay-ledger receipt classes because the record effect is suffix-dependent. Frozen-input transport reproduces its fixed-input control to numerical precision; changed later transitions, writes, and active caches place native omission outside that receipt class. Under the evaluated state/checkpoint interface, checkpoint replay reaches the record-omitted reference by recomputing the suffix.
2. **A training-free frozen-Gemma retrofit.** Prefix-mass preservation and one box per prefix solve adapt the companion support-vector mechanism to a pretrained hybrid-attention model. Unlike LoLCATs’ attention-transfer-plus-LoRA pipeline and Mamba-in-Llama’s stepwise distillation (Zhang et al., 2024; Wang et al., 2024), the retrofit keeps pretrained parameters fixed and performs per-prefix support-vector solves at prefill. At 4B, it matches base admission with 1.85% perplexity overhead.

Table 1: *Scope relative to the companion preprint (Ramesh, 2026).* The papers share a mechanism but test different model regimes and questions.

aspect	companion preprint	this paper
mechanism	introduces the support-vector gate, fixed- C solver, and reserve/decrement certificate	reuses the gate, solver, and certificate
model regime	compact language models trained from initialization	frozen Gemma 3 and inference-only native Kimi Linear
technical result	trainable support-vector attention	prefix-mass preservation, one C_b per prefix solve, and the native-KDA receipt/forcing audit
deletion claim	fixed- C retained-key refit in designed memory	training-free pretrained retrofit; failure of static/frozen receipts for native KDA omission

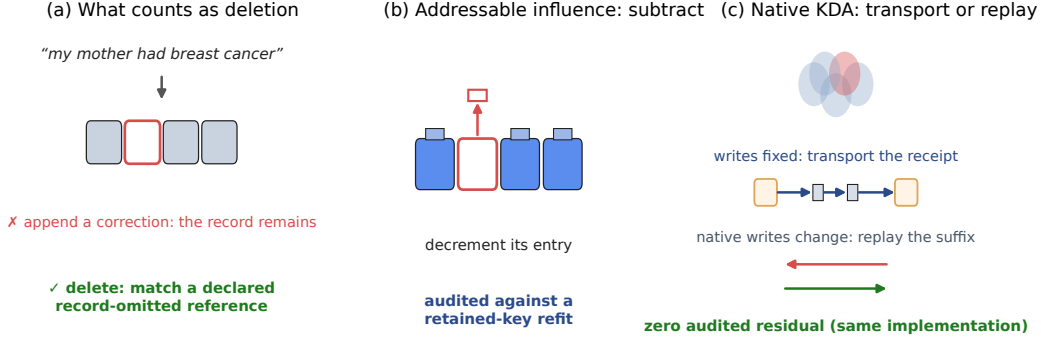


Figure 1: *The reference fixes the target; the representation constrains the available audited paths.* (a) Appending a correction leaves the old record in memory; deletion instead targets a declared record-omitted reference. (b) Addressable influence can be decremented: Gemma’s executed deletion state—verified decrement with exact-refit fallback—matches its retained-key refit within maximum KL 1.3×10^{-10} across 64 whole-record probes. (c) KDA’s later writes make a fixed receipt suffix-dependent (corpus-pooled raw range 12–49%). Full-transition transport is accurate to 6.1×10^{-6} in the frozen control. Changed transition/write forcing and active caches place native omission outside the tested receipt classes. Replay recomputes the suffix; zero residual through 18,842 tokens on final logits and all 80 audited KDA arrays verifies restoration across the declared checkpoint surface. Replacement replay supplies the corresponding amendment check.

2 RELATED WORK

Unlearning by design. MUNKEY deletes learned per-example keys from an external image-classification bank (Laguna et al., 2026). We apply its architectural premise to already-ingested autoregressive state; retraining and SISA reach the same counterfactual at a different substrate and cost (Cao & Yang, 2015; Bourtole et al., 2021).

Behavioral LLM unlearning. ICUL supplies the prompt-only inference-time baseline (Pawelczyk et al., 2024); TOFU, MUSE, and WMDP motivate attacks in their usual weight-edit setting (Maini et al., 2024; Shi et al., 2025; Li et al., 2024a). Relearning, representation, metric, and sampling critiques motivate our controls (Hu et al., 2024; Xu et al., 2026; Thaker et al., 2024; Lynch et al., 2024; Reisizadeh et al., 2025); our audit targets state/reference equality, which is stronger than behavioral suppression.

Grafting and cache editing. Trained conversion pipelines learn around the replacement: LoLCATs uses attention transfer followed by LoRA (Zhang et al., 2024; Hu et al., 2022), while Mamba-in-Llama uses attention-to-Mamba initialization and stepwise distillation (Wang et al., 2024). Our

Gemma retrofit freezes all pretrained parameters. The shared comparison axis is training requirement; downstream scores arise from different models and evaluations. H2O/SnapKV prune, VeriCache verifies against a retained full cache, and KVEraser learns an approximation to suffix recomputation (Zhang et al., 2023; Li et al., 2024b; Yao et al., 2026; Li et al., 2026a); our references state explicitly whether ingestion imprint remains.

Recurrent memories. Linear attention, delta networks, state-space models, and test-time memories compress history into recurrent state (Katharopoulos et al., 2020; Yang et al., 2024; 2025; Gu & Dao, 2024; Behrouz et al., 2025); Kimi Linear is hybrid (Zhang et al., 2025). Erase-then-Delta learns an erase address (Li et al., 2026b); our criterion adds certification against a named record-omitted reference. We bring deterministic replay (Abdullah, 2025) to inference state and measure the transport/changed-write boundary.

3 EXACT DELETION AS A REPRESENTATION PROBLEM

3.1 THE COUNTERFACTUAL STANDARD

Let $R = (r_1, \dots, r_n)$ be a sequence of records, $B(R)$ the persistent memory produced by ingesting them, and $f(q; B(R))$ the model’s next-token distribution for query q . Deleting r_j is exact at the output when

$$f(q; D_j(B(R))) = f(q; B(R_{\setminus j})), \quad (1)$$

up to a declared numerical tolerance. Equality of the memory states themselves is stronger and implies Equation 1. Establishing either equality requires a direct audit: behavioral suppression is compatible with a third state distinct from both sides of the equation. The units r_i and builder B are part of the declared reference: they are contextualized key records for Gemma’s retained-key certificate and raw input records for Kimi’s replay certificate.

The reference must also be named precisely. Gemma’s certificate holds the already-contextualized retained keys fixed and compares a decrement with refitting the support-vector solve on those keys; it therefore certifies the edited persistent store conditional on ingestion. A full token-stream rebuild also removes the way r_j shaped neighboring keys during ingestion. Kimi’s replay reaches that stronger reference; within one deterministic implementation, its audit reports zero float32 residual in final logits and every KDA cache array. We report two distinct standards: conditional retained-key equality for Gemma and native logical-state equality for Kimi.

Gemma’s conditional retained-key target provides a local audit path: a completed reverse update touches affected boundary solves rather than re-ingesting the raw context. In the corrected 4B audit, scoring the 32 already-constructed exact/refit/decay probe states totals 2,355 seconds; shared state construction is excluded. The current verifier constructs the retained-key refit before attempting decrement, and no corrected-configuration paired decrement-versus-full-repack timing benchmark exists. A separate legacy position-matched diagnostic measures the excluded ingestion imprint at 2.0×10^{-4} – 1.3×10^{-3} nats. The gap is small but nonzero and is not folded into the certificate.

3.2 STATIC ADDRESSES, TRANSPORTED RECEIPTS, AND REPLAY

Call a record *addressable* when stored bookkeeping a_j permits

$$\text{Dec}(B(R), a_j) = B(R_{\setminus j}) \quad (2)$$

without reprocessing other records. Addressability can be nonadditive: the stored solve and a_j need only recover the removal. A key–value entry is the simplest example (Laguna et al., 2026).

A recurrent write exposes the boundary. For prefix P , victim x , and suffix S , define the victim’s state contribution in common coordinates as

$$\delta_x(S) = B(P, x, S) - B(P, S). \quad (3)$$

A *receipt* is a saved value of this effect. Unchanged subtraction requires $\delta_x(S)$ to be suffix-independent; additive writes qualify, while diagonal decay requires a running product to *transport* the receipt through later updates.

Table 2: *Contract card*. Subtraction follows Ramesh (2026); equality is relative to the stated reference, while behavioral attacks assess extractability.

path	assumes	verified equal to	stops at
subtract	an address exists; fixed C ; contextualized keys	the retained-key refit, by solve identity and float64 output KL	one point in time; the neighbor imprint remains
transport	the later A_t, b_t are identical	the frozen-input recurrent matrix, by Proposition 1 and float32 relative residual	its fixed-input control; native forcing and active caches lie outside
replay	an exact boundary; a deterministic stored suffix	the native record-omitted logical hybrid state, by zero residual in all 21/21 qualified attempts (22 attempted)	one implementation; active state; costs $O(S)$

Operational characterization of the evaluated paths. These contracts define the two receipt classes tested here. Static deletion requires suffix-independent influence (Ramesh, 2026); frozen affine transport requires identical later numerical transitions and writes. A native trajectory lies outside a class when it violates that class’s premise. Methods that retain or recompute additional counterfactual suffix information expose a different interface; checkpoint replay is the recomputation path evaluated here.

Proposition 1 (Fixed-suffix transport identity). *Let $M_t = M_{t-1}A_t + b_t$. If present and omitted trajectories receive identical numerical (A_t, b_t) after boundary τ , then $\Delta_\tau = M_\tau^+ - M_\tau^-$ obeys*

$$\Delta_T = \Delta_\tau A_{\tau+1} \cdots A_T, \quad M_T^+ - \Delta_T = M_T^-, \quad (4)$$

where an empty product is I .

Proof. The common b_t cancels, giving $\Delta_t = \Delta_{t-1}A_t$; iterate. $\square$

If omission changes the later transition and write terms, denoted $+$ for record-present and $-$ for record-omitted, direct subtraction gives

$$\Delta_t = \Delta_{t-1}A_t^+ + M_{t-1}^-(A_t^+ - A_t^-) + (b_t^+ - b_t^-). \quad (5)$$

The first term is the transported boundary receipt; the remaining terms are the exact transition and write forcing omitted by frozen-input transport. Their measured magnitude therefore tests the transport premise directly.

Example 1 (Native omission changes later writes). KDA has exactly this affine form for fixed kernel inputs: $A_t = \text{diag}(g_t)(I - \beta_t k_t k_t^\top)$ and $b_t = \beta_t v_t k_t^\top$. In measured 48B present/omitted runs, median relative suffix-input differences are 0.150/0.246/0.025/0.088 for $k/v/g/\beta$, and the median forcing-to-difference norm ratio is 0.84. Convolution and attention caches also change. The measured native KDA trajectories consequently lie outside Proposition 1’s premise; suffix-invariant affine updates remain exactly transportable under the proposition.

Replay implementation oracle. Let deterministic builder B produce the complete logical active state and write $R = P\|x\|S$. Exact restoration of $B(P)$, followed by the stored S under the same model, tokens, arithmetic, and execution order, gives, by induction over S ,

$$\text{Replay}(\text{ckpt}(B(P)), S) = B(P\|S) = B(R_{\setminus x}).$$

We use this consequence as an implementation oracle. Zero residual on final logits and every audited KDA cache array—three convolution arrays and one recurrent matrix in each of 20 layers—together with restored MLA offsets verifies replay across the declared checkpoint surface. Here S is the stored input suffix; inactive backing-buffer sanitization and cross-hardware reproducibility remain separate requirements.

3.3 ADDRESSABLE REALIZATION: THE SUPPORT-VECTOR GATE

The implementation uses the support-vector solve of Ramesh (2026); we recap the definitions used by the retrofit. Given context keys $x_i \in \mathbb{R}^d$, values v_i , and a query q , the gate reads out a kernel-weighted

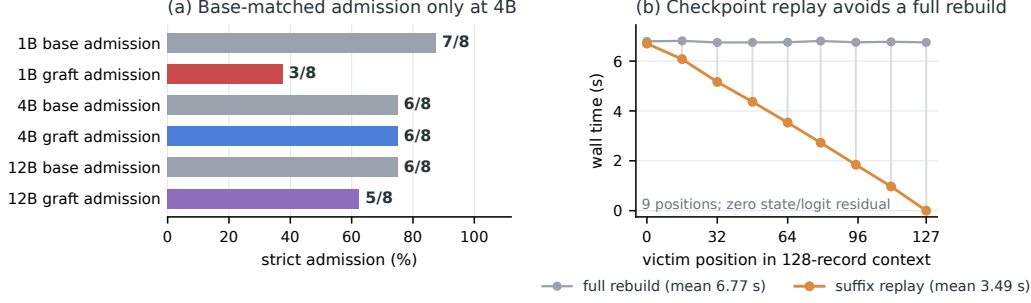


Figure 2: *Feasibility, recall, and recomputation are different questions.* (a) On the untouched cohort, base→graft admission is $7/8 \rightarrow 3/8$ at 1B, $6/8 \rightarrow 6/8$ at 4B, and $6/8 \rightarrow 5/8$ at 12B; all eight records are fixed- C_b -feasible at every scale. The 4B graft also passes field rank/lift 23/24 each and all 8/8 retained neighbors. (b) Kimi suffix replay and full rebuild reach the same logical record-omitted target with zero audited residual. Across nine sampled victim positions in one 128-record, 3,248-token context, paired means are 3.487 versus 6.771 s; each point is one timing, not a repeated-run confidence interval.

average whose weights are gated by coefficients α solving a one-class support vector description (Tax & Duin, 2004) over the keys. Each chunk boundary is its own solve over n_b prefix keys, so its box is $C_b = 1/(\nu n_b)$, with $\nu \in (0, 1]$. The solve partitions the keys into margin, error, and *reserve* sets, the last with coefficients identically zero. Because $\sum_i \alpha_i = 1$, raw coefficients would make the whole prefix fade as it grows; we rescale the surviving gated prefix back to its pre-gate attention mass. Zeros remain zero. The **certificate** relies on two properties: a reserve token contributes exactly zero to the readout, and a completed reverse C&P path returns to the retained-key optimum (Cauwenberghs & Poggio, 2000); failed paths use exact refit. For every deletion certificate, each C_b is frozen at its pre-deletion value; “retained-key refit” therefore uses that fixed C_b rather than recomputing the box from unchanged ν . The certificate compares the float64 decrement (or exact-refit fallback) with this float64 reference. “Masked refit” below is a finite-iteration FP32 drop-position refit used only for behavioral tests; “decay,” our approximate foil, scales a coefficient toward zero and leaves a residual. Feasibility after deleting n_{del} keys is $n_{\text{del}}/n_b \leq 1 - \nu$ at every affected boundary. We check this before editing and route violations to disclosed full repack. The contract applies to a sealed point-in-time solve: a later key insertion can make an old reserve point active. The companion preprint’s deterministic counterexample motivates the prefill-once audit, which seals each solved boundary and scopes certification to that boundary (Ramesh, 2026).

Repacking the raw context without X and re-ingesting reaches the stronger, imprint-free record-omitted reference. Decrement instead reaches the fixed- C_b retained-key target; full repack remains the honest fallback when ingestion imprint must also be removed. Figure 2b gives the artifact-backed replay/rebuild comparison for the case in which both paths reach the same native target.

4 TRAINING-FREE GEMMA INSTANTIATION

Graft. Gemma 3 (Gemma Team, Google DeepMind, 2025) interleaves five local sliding-window layers for every global layer. We replace only the global layers (four at 1B, five at 4B, and eight at 12B) and reuse their projections, normalization, rotary embeddings, and grouped-query layout. The configuration is frozen at $\nu = 0.7$, chunk length 128, deterministic solver seed 0, and a deterministic 256-key median bandwidth. The retrofit keeps pretrained weights fixed and adds per-prefix support-vector solves at prefill.

Prefill-once protocol and admission. The audit ingests memory once, snapshots native caches plus every boundary’s keys, values, coefficients, bandwidth, and C_b , then seals the boundary. Queries fork this state and never reinsert the record. The final confirmation manifest was frozen after recipe selection and contains eight untouched three-field TOFU records (Maini et al., 2024). Strict admission requires all three fields, one unrelated retained neighbor, and fixed- C_b feasibility. At 4B, both ungrafted and grafted models admit the same 6/8 records: field rank passes 23/24, field

Table 3: *The corrected graft is training-free.* WikiText-103 perplexity on 400 fixed blocks; admission is the untouched eight-record confirmation (ungrafted $\rightarrow$ graft).

model	ungrafted PPL	graft PPL	cost	strict admission
Gemma-3-1B-PT	26.70	27.69	+3.70%	7/8 $\rightarrow$ 3/8
Gemma-3-4B-PT	21.56	21.96	+1.85%	6/8 $\rightarrow$ 6/8
Gemma-3-12B-PT	22.20	24.80	+11.70%	6/8 $\rightarrow$ 5/8

lift 23/24, retained rank 8/8, and fixed- C_b feasibility 8/8. At 1B, the graft admits 3/8 versus 7/8 ungrafted (rank 20/24, lift 24/24, retained 7/8); at the predeclared 12B second checkpoint, it admits 5/8 versus 6/8 ungrafted, with all eight fixed- C_b -feasible. The additional 12B graft rejection arises from one target-field rank criterion. Behavioral evaluation centers on 4B, the scale that preserves base admission; the 1B and 12B outcomes document checkpoint-specific retrieval/readout boundaries.

Numerical certificate. Every confirmation record is certified regardless of recall admission: eight records and 32 full-vocabulary probes at both 1B and 4B. Float64 decrement and retained-key refit use each boundary’s identical frozen C_b , and each state computes its own mass-preserving rescale. At 1B, median/max KL are $1.86 \times 10^{-13}/1.25 \times 10^{-10}$; at 4B they are $2.36 \times 10^{-12}/6.48 \times 10^{-11}$. All 64 probes remain below 10^{-6} . The reverse path uses exact refit after failed post-verification in 396/640 1B and 2,020/3,040 4B attempted head-boundary decrements. That fallback is the retained-key refit itself, so it changes what the audit costs and not what it certifies: the KL values above already report the executed path. Denominators count only boundaries holding a deleted position; the enumerated gate counts (1,024 and 4,320) also include boundaries that attempt no decrement. The verifier constructs refit before decrement, so its corrected timing characterizes the joint audit; Gemma speedup remains unmeasured. A same-key 4B diagnostic isolates the mechanism: as ν moves $0.3 \rightarrow 0.5 \rightarrow 0.7$, mean support fraction rises $30.7\% \rightarrow 50.5\% \rightarrow 70.3\%$, margin fraction falls $1.39\% \rightarrow 0.97\% \rightarrow 0.64\%$, and reverse fallback rises $0\% \rightarrow 7\% \rightarrow 59\%$. More retained support therefore trades against incremental margin headroom. The predeclared 12B extension reports admission and quality; decrement/refit certification covers 1B and 4B.

Quality. Table 3 uses the same 400 packed WikiText-103 token blocks at all three scales. The independent model unit is one checkpoint per scale; the 20-block groups describe corpus heterogeneity. Ungrafted 12B perplexity 22.20 exceeds 4B’s 21.56 by 3.0% on this fixed packing; the 12B value reproduces under a second loss implementation on the same token tensor, while alternative packings, backends, and corpora remain untested and official Gemma documentation reports no size-stratified WikiText perplexity. The +11.70% graft cost is paired within the 12B checkpoint and therefore does not depend on cross-scale base comparability. Costs are +3.70%, +1.85%, and +11.70% at 1B, 4B, and 12B. For descriptive heterogeneity, twenty consecutive 20-block groups give median/IQR/range $3.52/[3.17, 4.16]/[2.31, 6.17]\%$ at 1B, $1.61/[1.32, 2.40]/[0.50, 3.78]\%$ at 4B, and $11.46/[9.35, 13.10]/[6.82, 23.41]\%$ at 12B. Every 12B group is positive. These groups are not independent model replications, so we report no inferential interval.

Behavioral attacks on the corrected configuration. For each of the six admitted 4B records, we sample 200 completions for three field prompts and one composite prompt under present, masked-refit, decay, prompt-only ICUL, and never-stored conditions. Field Leak@200 is 100% for present/ICUL, 27.8% for decay, and 16.7% for both masked refit and never stored; the paired record-level difference is exactly 0.0 [0.0, 0.0] (10,000 bootstrap draws). The same three field queries leak in both conditions: two have never-stored first-token ranks 3 and 6, and the third is a low-probability rank-94 floor event, locating all three observations at the never-stored floor. Composite Leak@200 is likewise 16.7% for both. Under 0–8 target-free elicitation shots, masked-refit mean normalized recovery stays 0.009–0.011, while ICUL stays near 1. Related-data LoRA updates through budget 256 leave recovery near zero. On the final six records, LiRA gives masked-refit AUC 0.534. To increase breadth, a separately frozen cohort attempts all 20 TOFU author blocks, admits 16, and evaluates 512 positive and 512 negative draws: masked-refit/full-repack policy AUC is 0.517, with TPR 1.4%, 6.1%, and 10.4% at FPR 1%, 5%, and 10%; present AUC is 0.99994, ICUL 1.0, and decay 0.556. All six-record attacks use masked refit throughout; the broad LiRA policy routes 16/512 masked-refit test deletions (3.1%) to full repack under the fixed- C_b precheck, and the reported AUC includes them.

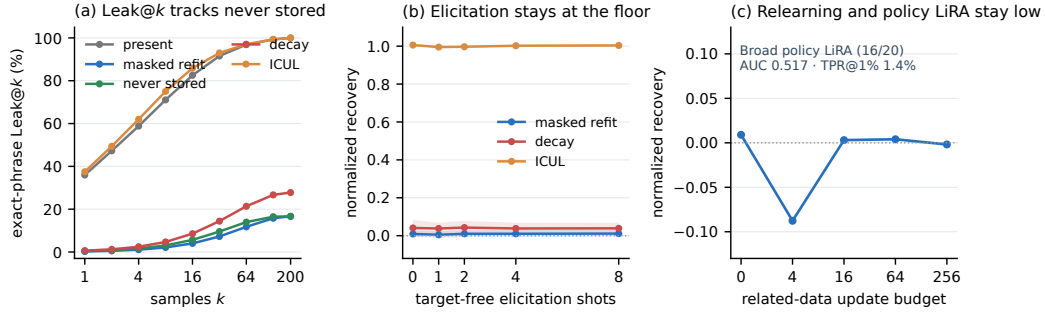


Figure 3: *Corrected-config behavioral attacks at 4B.* (a) On the six admitted confirmation records (18 field queries), masked-refit and never-stored Leak@200 both equal 16.7%, while present and prompt-only ICUL reach 100%. (b) Six-record means and 95% record-bootstrap intervals keep masked refit near the never-stored floor through eight target-free shots. (c) The six-record related-data curve stays near zero through 256 updates; the annotation instead reports the separately frozen broad LiRA cohort (16/20 admitted), where the masked-refit/full-repack policy is near chance. These panels report behavioral attack evidence; the float64 certificate is evaluated separately.

Arbitrary reassignment of all 16 routed scores gives the deterministic aggregate-AUC sensitivity range $[0.486, 0.549]$, a maximum movement of $0.03125 = 16/512$; this range is not a confidence interval. Behavioral attacks assess extractability; the float64 audit establishes reference equality.

Legacy stress tests. Appendix A.1 quarantines the earlier trained-conversion development path as legacy-only diagnostics tied to its distinct readout, training procedure, and comparator. Comparative claims use the training-free configuration.

5 NATIVE KDA RULES OUT FROZEN RECEIPTS

The released 48B Kimi Linear hybrid provides a native test of the receipt premises in inference-only mode with all parameters fixed at their released values (Zhang et al., 2025; DeepSeek-AI, 2024; Yang et al., 2024; 2025).

Hybrid state requires both channels. Setting the recurrent matrix in all 20 KDA caches to zero ($\gamma = 0$) leaves the planted ward code at rank 1 ($p = 0.881$). The seven MLA layers continue to hold the text verbatim, while the three short-convolution arrays in each KDA cache remain unchanged; the intervention therefore shows that recurrent-state zeroing alone leaves a live hybrid-memory path (Appendix Table 8).

Static and decay-ledger receipts remain suffix-dependent. Across three corpora, one record’s KDA contribution changes by 12–49% across suffixes (pooled raw extrema 0.116–0.489); applying the diagonal-decay ledger still leaves 8–49% (corrected extrema 0.081–0.494). Native KDA therefore has a suffix-dependent record effect before and after decay-ledger correction.

Table 4: *Native KDA remains suffix-dependent after a decay ledger.* Corpus-pooled KDA extrema, rounded to percentage points, are 0.116–0.489 $\rightarrow$ 12–49% raw and 0.081–0.494 $\rightarrow$ 8–49% corrected; the slightly larger corrected maximum shows that the ledger can increase suffix dependence.

Write rule	Synthetic		MIMIC-CDS		MIMIC-notes	
	raw	corr.	raw	corr.	raw	corr.
Additive writes only	$\leq 3.1e-6$	—	$\leq 2.1e-6$	—	$\leq 1.4e-5$	—
+ per-channel decay	.007–.066	$\leq 7.0e-6$	.008–.078	$\leq 6.4e-6$	.015–.109	$\leq 7.0e-5$
+ delta rule (= KDA)	.151–.419	.139–.418	.116–.300	.089–.300	.122–.489	.081–.494
End-to-end (full model)	1.02 (.38–1.24)		0.95 (.21–1.23)		0.81 (.28–1.16)	

Frozen transport isolates the fixed-input boundary. Propagating the boundary difference through a fixed sequence of affine KDA updates achieves 6.1×10^{-6} maximum relative error. Native omission, however, changes the later transition and write inputs. In six source-free configurations, the forcing decomposition is satisfied to numerical precision and the forcing-to-difference norm ratio has median 0.84; convolution and MLA cache state also change. Together, those inputs place the native omitted state outside the frozen-receipt path.

Checkpoint replay reaches native omission by recomputing the suffix. The executor restores the pre-record logical checkpoint and reprocesses the surviving suffix. Zero residual against the never-ingested final logits and all 80 audited KDA cache arrays verifies replay across the declared checkpoint surface within the deterministic implementation. Timing, storage, clinical denominators, zero-residual audits, and amendment are reported in Appendix G.

Direct-channel removal is a behavioral comparator. Attention-only masking disagrees with replay on every tested victim, placing direct-channel suppression in a state distinct from native omission (Appendix Tables 7 and 9).

6 LIMITATIONS

- **Scale and generality.** Among the three tested checkpoints, 4B uniquely preserves base admission with low overhead. The predeclared 12B extension is fully fixed- C_b -feasible and its admission is one record short of base (5/8 versus 6/8); the binding failure under this protocol is the +11.70% quality cost. Outcomes across the three checkpoints are nonmonotone. Behavioral attacks condition on 4B recall admission; the broad LiRA cohort admits 16/20 attempted records from one TOFU packing rule and one checkpoint. Cross-family and cross-hardware replication remain open. Muse Glimmer (Meta AI, 2026) is a concrete cross-family candidate that replaces global RoPE with NoPE, adds an attention-output gate, and uses 16:1 GQA; a controlled port must resolve all three, so evaluation remains future work. Kimi likewise uses one deterministic implementation/workstation.
- **Scope.** Our deletion target is in-context state; pretrained weights and prior outputs remain unchanged (Wu et al., 2025). Gemma’s retained-key certificate leaves neighbor imprint, which repack removes. Executor coverage comprises the grafted long-range store; local-window text and inactive slots remain outside it. The certificate applies to the sealed point-in-time solve. Records exceeding $1 - \nu = 30\%$ of an affected boundary route to full repack. Within the eligibility bound, post-verification routes 61.9% of 1B and 66.5% of 4B attempted boundary decrements to exact refit; corrected latency advantage remains unmeasured. The support-vector solve is also heavier than softmax.
- **KDA scope and recomputation cost.** The measured native trajectories place exact native omission outside the static, decay-ledger, and frozen-input receipt classes. Redesigned representations and algorithms that record additional counterfactual suffix information define a broader interface. The executed exact path is checkpoint replay; it stores 41.4 MiB per boundary, with sparse-checkpoint latencies estimated.

Future work. Future work should test segmented, record-wise ingestion for imprint-free writes and derive streaming certificates that remain valid under later admissions. The opposite 1B/12B failure modes suggest separate mechanisms: readout fidelity at 1B versus cumulative quality cost from eight gated global layers at 12B compared with five at 4B. Test this layer-count hypothesis with matched 4/5/8-layer subsets at 12B while holding packing, ν , bandwidth, and solver seed fixed; also test legacy proper-name sensitivity. For KDA, richer transport or compressed replay must account for input-dependent transitions, writes, and hybrid cache state while preserving a checkable reference.

Conclusion. Native KDA and frozen Gemma expose opposite sides of the same design question. Omitting the record changes the later transitions and writes, so a fixed receipt cannot be carried through them; that places native omission outside the receipt classes we tested. Checkpoint replay verifies restoration across final logits and all 80 audited KDA arrays. At 4B, prefix-mass preservation and per-prefix boxes clear the tested admission-and-utility bar: base-matched 6/8 admission with 1.85% perplexity overhead through a training-free retrofit.

ETHICS STATEMENT

This work aims to make deletion from a model’s persistent memory auditable, in direct support of erasure rights (e.g. GDPR Article 17), clinical consent withdrawal, and the removal of sensitive records from assistant memory. The forgetting benchmarks use no real personal data: every forgotten “fact” is a fictitious TOFU biography or a synthetic benchmark record, and the patient and incident scenarios are invented. The hybrid study (Section 5) additionally evaluates deletion on real clinical records from MIMIC-IV-Ext-CDS and MIMIC-IV-Note, used under PhysioNet credentialed access and its data use agreement: all processing is local, reports are aggregate-only, no note text, identifier, or timestamp is read beyond the evaluated fields, no generations are produced from record-bearing contexts, and a substring audit over all source values runs before any report is written. No MIMIC data is redistributed. We report the method’s boundaries as claims of equal standing with its strengths—the ingestion-time imprint on neighboring representations, the pre-deletion-snapshot adversary that no current method defeats, and the scale-dependence of utility and certificate precision—because an overstated deletion guarantee is itself a privacy harm. Because certified deletion can remove evidence of provenance or safety-relevant content, deployment requires authenticated authorization and tamper-evident logging. The audits establish agreement with their declared references at audit time; the logging layer establishes that an authorized deletion event occurred.

REPRODUCIBILITY STATEMENT

The training-free graft, per-boundary boxes, mass-preserving readout, float64 decrement, and every evaluation are released as code. The Gemma study pins model and dataset revisions, solver seed, bandwidth rule, configuration manifest, and untouched confirmation cohort; source-state and report hashes bind every headline value. Experiments run on one Apple-silicon machine. The behavioral substrate is public TOFU (Maini et al., 2024), and quality uses fixed WikiText-103 blocks (Merity et al., 2017). We release the full pipeline, including the sharded robustness runner and the scripts that regenerate every figure from recorded artifacts (claim-to-command mapping in the repository’s provenance file). The base model is the publicly available Gemma 3 family (Gemma Team, Google DeepMind, 2025), used under its license. The hybrid study runs inference-only on the released 8-bit Kimi Linear weights under MLX on the same machine; its synthetic-record variants of both headline measurements reproduce without credentialed data, and the MIMIC runners take an explicit local path and emit the aggregate-only JSON reports the tables are read from. The introduction’s amendment scenario is reported as an aggregate audit with an invented transcript in Appendix G. Source, artifact, and claim-to-command release details are provided in the accompanying public reproducibility package.

AI USE STATEMENT

We used generative-AI tools for literature discovery, hypothesis refinement, experimental and methodological feedback, code implementation and refactoring, test design, figure and manuscript drafting, and copy editing. Every AI-assisted claim, citation, equation, experiment, and artifact was reviewed by the author; executable claims were checked with committed tests or recorded experiment outputs, and citations were checked against primary sources. No credentialed MIMIC record or source text was provided to a generative-AI service. The author takes responsibility for the final content of the paper and all accompanying artifacts.

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A EXPERIMENTAL DETAILS

Grafting a certified support-vector memory into a deployed LLM

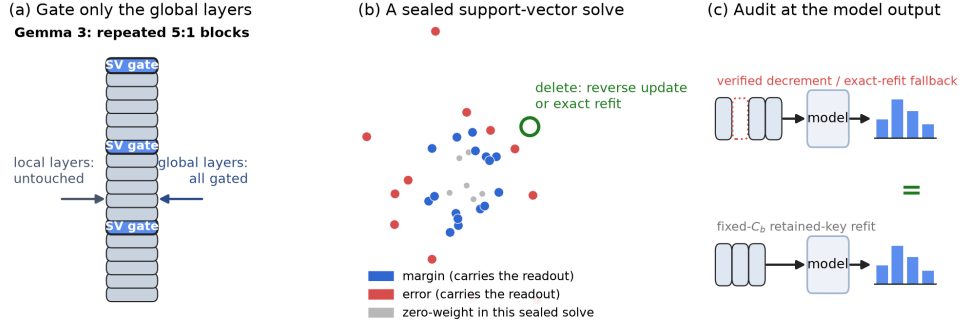


Figure 4: *Where the addressable memory lives, and how a fact leaves it.* (a) Gemma 3 interleaves five local layers per global layer; only the global layers are replaced. (b) The companion one-class support-vector fit (Ramesh, 2026) over the keys assigns exact zeros and supports a reverse incremental update. (c) Deletion is audited at the output: the decremented memory and a fixed- C_b retained-key refit drive the live model in double precision and must produce the same next-token distribution. The fact sits beyond the local sliding window, so the gated memory is its only carrier.

Models and training-free graft. We evaluate pinned google/gemma-3-1b-pt, google/gemma-3-4b-pt, and google/gemma-3-12b-pt revisions. Under Gemma 3’s 5:1 interleave, the 1B model has four global layers (5, 11, 17, 23), 4B has five (5, 11, 17, 23, 29), and 12B has eight (5, 11, 17, 23, 29, 35, 41, 47); only these layers are replaced. The graft reuses every projection, query-key normalization, rotary embedding, and grouped-query layout. It uses $\nu = 0.7$, chunk length 128, and $C_b = 1/(\nu n_b)$ for each boundary’s n_b -key solve. After applying the coefficients, the long-range prefix is rescaled to its pre-gate softmax mass per query. Reserve coefficients remain exactly zero.

There is no attention-transfer stage, LoRA, FineWeb continuation, or language-model parameter training; per-prefix support-vector solves remain part of prefill. The bandwidth is the median key distance over a deterministic 256-point stratified sample and is frozen at prefill. MLX FISTA uses solver seed 0; repeat runs produce identical admission payloads. The source-free

configuration is `benchmarks/mass_preserving_boundary_v1.json`, and the compact evidence aggregate is `benchmarks/iclr_mass_preserving_boundary_v2.json`. The predeclared 12B extension and source-free result are `benchmarks/training_free_second_checkpoint_{v1,result_v1}.json`.

Frozen confirmation. Recipe development used the earlier synthetic, paired, and continuation manifests. The final `whole_record_confirm_v2.json` cohort was then frozen: it exhausts the four unused 20-row TOFU blocks and selects two disjoint, predetermined three-field groups per block. All eight outcomes remain in the denominator. Quality uses the same 400 packed WikiText-103 token blocks at all three scales.

Evaluation. Perplexity is WikiText-103 test on 400 fixed blocks. Admission reports all eight confirmation records and requires three target fields, one retained neighbor, and feasibility at every affected boundary. The float64 certificate covers all eight confirmation records at 1B and 4B, independent of recall admission, with three field probes plus one composite probe per record. The 12B extension covers admission and quality only. Corrected-config behavioral attacks use the six admitted 4B records. Decay scales target coefficients by $\gamma = 0.01$. Earlier recovered-model panels remain below as legacy diagnostics.

Prefill-once audit. The stateful audit executes the stored prefix once with native Hugging Face caching enabled and simultaneously snapshots each grafted layer’s keys, values, solved boundary gates, boxes C_b , and kernel bandwidth. The long-range boundary is then sealed: query tokens remain on the local continuation path and are not inserted into the stored solve. Deletion recomputes the saved boundary gates once, either from float64 decrement/refit coefficients or the single-precision masked solve. Each audit prompt receives a deep fork of the same edited cache. Regression tests fail if the legacy token-ID recomputation methods are called. Before an edit, each affected boundary checks $n_{\text{del}}/n_b \leq 1 - \nu = 0.3$. Violations are machine-readably routed to full repack; all eight final-confirmation records pass at 1B, 4B, and 12B.

Admission decomposition. At 4B, ungrafted and grafted models reject the same two records for the same base-model reasons: one field fails the lift gate and one fails first-token rank. Thus both admit 6/8; the graft passes rank on 23/24 fields, lift on 23/24, the retained-neighbor gate on 8/8, and fixed- C_b feasibility on 8/8. At 1B, ungrafted admission is 7/8 but grafted is 3/8: field rank passes 20/24, lift 24/24, retained rank 7/8, and feasibility 8/8. These are conjunctions, not post-hoc subsets. The exact frozen 1B operator gives 7/8 on the shorter synthetic development manifest (64–80 deleted positions), 3/8 on paired TOFU, 4/8 on the first untouched TOFU continuation, and 3/8 on final confirmation. Thus the earlier reported 8/8 used the pre-correction global- C operator and is not a like-for-like development score. Across the final cohort all field lifts pass; the extra 1B failures are four target-token ranks and one retained rank. Both scales use 256-dimensional heads, so “narrower keys” is not an explanation; 1B instead has fewer global layers (4 vs. 5), attention heads (4 vs. 8), and KV heads (1 vs. 4), without establishing which difference is causal.

Predeclared 12B second checkpoint. The unchanged recipe and final confirmation cohort were frozen before loading the cached 12B-PT checkpoint. Ungrafted admission is 6/8; the graft reaches 5/8, with fixed- C_b feasibility 8/8. The two arms share one secret-lift rejection and one rank rejection; the graft adds one target-field rank failure. Ungrafted/grafted field-rank passes are 23/24 and 22/24; answer-lift, secret-lift, and retained-neighbor passes are respectively 24/24, 23/24, and 8/8 in both arms. All outcomes remain in the source-free result.

Certificate decomposition. Each certified scale (1B and 4B) contributes eight records and 32 probes, including recall-rejected records. At 1B, exact/refit median and maximum KL are 1.86×10^{-13} and 1.25×10^{-10} ; at 4B they are 2.36×10^{-12} and 6.48×10^{-11} . No probe exceeds 10^{-6} . Decay/refit median (maximum) is 0.048 (3.09) at 1B and 0.0148 (0.588) at 4B. Reverse decrement falls back to exact refit in 396/640 1B (61.9%) and 2,020/3,040 4B (66.5%) attempted head-boundary decrements. Denominators exclude the 384 1B and 1,280 4B enumerated gates that precede every deleted position and so attempt no decrement. Full-forward and cached-decode tests independently verify that exact and refit states compute their own mass-preserving rescale from coefficients obtained at the same boundary box.

To isolate the fallback increase, a frozen one-record diagnostic reuses identical 4B contextualized keys, deleted positions, per-layer bandwidths, and post-verification tolerance while changing only ν . At $\nu = .3, .5, .7$, mean support fractions are 30.7%, 50.5%, 70.3%; margin fractions are 1.39%, 0.97%, 0.64%; and reverse fallback is 0/400, 28/400, 236/400. Thus $\nu = .7$'s greater retention coverage produces larger support sets and thinner margin sets, making reverse incremental paths less robust. These denominators also count attempted decrements, so the $\nu = .7$ rate of 59% is directly comparable to the cohort rates above. This controlled diagnostic explains the direction of the all-record rate; it is not a second cohort estimate.

Corrected audit-cost scope. The versioned aggregate `benchmarks/corrected_audit_cost_v1.json` derives probe timers from the hash-bound corrected certificate reports. Scoring already-constructed exact, retained-key-refit, and decay states totals 561.5 seconds over 32 probes at 1B and 2,354.6 seconds over 32 probes at 4B. These timers exclude model loading, prefill, state construction, decrement/refit construction, and full repack. They characterize audit scoring cost, not deletion latency or speedup; no corrected paired decrement-versus-repack timing exists.

Corrected-config behavioral attacks. The six admitted 4B records yield 18 field queries and six composite queries. At $k = 1$, exact-phrase leak is 0.28% under masked refit and 0.42% when never stored. At $k = 200$, both are 16.7%; their paired record-bootstrap difference is 0.0 [0.0, 0.0]. Present and ICUL are 100%, and decay is 27.8%. Composite masked-refit and never-stored rates also coincide at 16.7%. The same (manifest, field) queries (0, 1), (3, 1), (5, 2) leak in both conditions; their never-stored first-token ranks are 94, 3, 6, respectively. Thus the equality is the measured guessable/chance floor, not three different residual-memory hits. Ten-thousand record-bootstrap draws are descriptive over $n = 6$, not a population claim.

For elicitation, record-mean normalized recovery under masked refit is 0.009 [−0.006, 0.026] at zero shots and 0.011 [−0.004, 0.035] at eight shots (95% record bootstrap); ICUL remains near one. Rerearning means at budgets 0, 4, 16, 64, 256 are 0.009, −0.088, 0.003, 0.004, −0.002. LiRA fits 32 in/out shadows per record and evaluates 32 held-out draws per record. On the final six-record cohort (192 per side), masked-refit AUC is 0.534. A separately frozen broad cohort takes the same field offsets from all 20 TOFU author blocks; strict admission passes 16/20, yielding 512 draws per side. The masked-refit/full-repack policy AUC is 0.517, with TPR 1.4%, 6.1%, and 10.4% at FPR 1%, 5%, and 10%. Present AUC is 0.99994, ICUL 1.0, and decay 0.556. All four broad-cohort rejections and their reasons remain in the admission report. No six-record attack invokes full repack. In the broad LiRA run, 16/512 masked-refit test deletions (3.1%) route to full repack under the fixed- C_b precheck; the reported AUC includes them. One positive score contributes at most 1/512 of AUC, so changing all 16 fallback scores arbitrarily can move AUC by at most 0.03125, giving the deterministic sensitivity range [0.486, 0.549]. This is not a confidence interval.

Quality dispersion. The aggregate paired costs $100[\exp(\overline{\Delta\text{NLL}}) - 1]$ are 3.704%, 1.851%, and 11.699% at 1B, 4B, and 12B. Twenty fixed consecutive groups of 20 blocks give median/IQR/range 3.52/[3.17, 4.16]/[2.31, 6.17]% at 1B, 1.61/[1.32, 2.40]/[0.50, 3.78]% at 4B, and 11.46/[9.35, 13.10]/[6.82, 23.41]% at 12B; every group is positive. The groups describe heterogeneity along one fixed corpus sequence. They are not independent checkpoints or IID replicates, so we report no confidence interval or p -value.

A.1 LEGACY RECOVERED-READOUT DIAGNOSTICS

Everything through the remainder of this Gemma appendix section predates the mass-preserving, boundary-local configuration. It used a prefix-shrinking $\nu = 0.3$ readout plus attention transfer and LoRA recovery. We retain these artifacts to document the development path and attack implementations, not as support for the headline model.

Proxy objective and timing scope. The released legacy audit's 4.3×10^{-3} proxy/exact geometric-mean KL does not isolate numerical precision: its float64 path selected a fresh per-head, per-boundary median bandwidth, whereas the FP32 path used the prefill-frozen `SVDecodeSession` bandwidth. We therefore mark that number objective-mismatched. The corrected path passes the identical frozen bandwidth and box to both solvers, evaluates the feasible projected FISTA iterate directly rather than a ridge-regularized partition reconstruction, and leaves the source session immutable. Likewise,

the plotted ~ 47 -s exact point includes the shared update that constructs and audits decrement and fixed- C refit states; it is not pure decrement latency.

Locked proxy frontier. After objective alignment, calibration indices $\{0, 2, 4, 6\}$ across three adapters sweep FISTA iterations $\{20, 40, 80, 160, 320, 640, 1280\}$, partition cutoffs $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, and a predeclared residual-cutoff grid. The fastest nontrivial zero-violation policy uses 160 iterations, partition cutoff 3×10^{-4} , and residual cutoff 3×10^{-3} : calibration has 1/12 fallback and max $\text{KL}(\text{refit64}||\text{hybrid}) = 4.27 \times 10^{-6}$. Evaluated once on locked indices $\{1, 3, 5, 7\}$, it has no 10^{-3} -gate violations, 2/12 exact-refit fallbacks, max KL 4.88×10^{-6} , mean absolute retained drift 5.55×10^{-4} nats, and 17.98 s mean end-to-end time. We call this a residual-qualified approximation, not a certificate.

Controlled multi-needle benchmark. At pinned RULER commit (Hsieh et al., 2024) ab17b785...ee13, eight seed-271828, 1,024-token contexts use the official niah_multitkey_1 complexity: four word keys, one seven-digit value per key, and one query. Our declared erasure adaptation adds one later conflicting value for the queried key as the owned span; deleting it restores the earlier gold needle. The source-free manifest (4a64a547...2e31) is fixed before model evaluation. Target admission passes 4/8, retained-key availability 8/8, and joint admission 4/8, without dropping the four failures. The all-eight, one-repeat method matrix remains a mechanism diagnostic. Full repack takes 2.68 s and defines zero behavioral KL; exact decrement, FP32 proxy, cache shift, decay, and correction-style ICUL have mean KL to repack 0.0144, 0.0146, 0.211, 0.0225, and 1.434, respectively. Exact and refit output distributions agree to maximum KL 9.51×10^{-12} ; the executed exact path uses 44/640 disclosed refit fallbacks. The exact row’s 59.12 s again includes shared decrement/refit construction, while the proxy takes 3.50 s. Every method completes all eight records and the source state remains immutable.

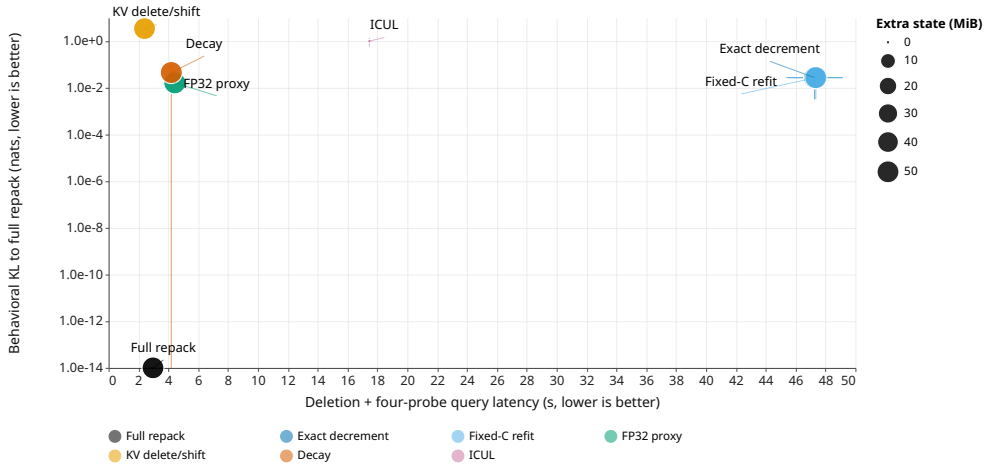


Figure 5: *Legacy recovered-readout frontier.* Means over all eight fixed records and three recovered seeds; rules are seed-level 95% Student- t intervals. Behavioral KL to full repack is vertical, deletion-plus-four-probe latency is horizontal, and point area is incremental state storage. Only 1/24 seed-record contexts passes the unchanged joint target-plus-retained recall gate (target-only 8/24, deleted fields 44/72, retained neighbor 8/24); fixed- C feasibility is 24/24 contexts and 120/120 boundaries. Thus this all-predeclared frontier is a mechanism diagnostic, not an efficacy estimate. Exact decrement targets the retained-key refit, which differs from full repack by ingestion imprint. KVERaser is excluded as technically incompatible, not plotted as missing data.

B LEGACY 1B RECOVERY UTILITY

These values diagnose the superseded two-stage recovery recipe; they are not the utility result for the training-free main configuration. The independent unit is the complete training seed, not a WikiText block or a certificate target. For each of seeds 0, 1, 2, the recovered and control models use identical stage-2 examples and LoRA initialization and are evaluated on the same 400 packed WikiText blocks. We first compute the paired per-seed utility cost, then report its arithmetic mean and two-sided 95% Student- t interval (two degrees of freedom); arm-level perplexity means and intervals are reported separately.

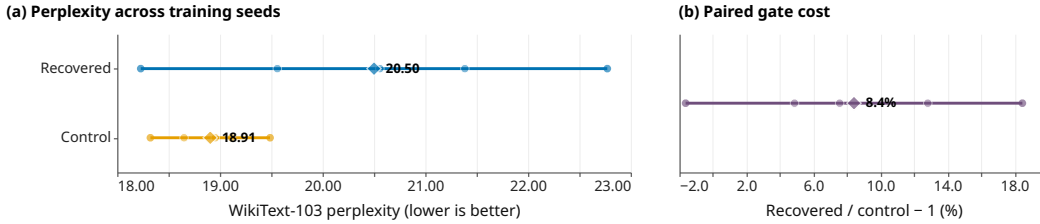


Figure 6: *Legacy stream-matched recovered-readout estimate*. Dots are the three exactly stream-matched training seeds, diamonds are means, and lines are two-sided 95% Student- t intervals. Recovered perplexity is 20.50 [18.23, 22.77], control is 18.91 [18.32, 19.49], and paired gate cost is +8.42% [-1.63, 18.47].

Table 5: *Every 1B control uses its recovery arm’s exact stage-2 examples*. WikiText-103 perplexity, 400 blocks.

seed	recovered	control	paired cost
0	21.39	18.95	+12.82%
1	20.56	19.11	+7.56%
2	19.56	18.65	+4.87%
mean	20.50	18.91	+8.42%
95% CI	[18.23, 22.77]	[18.32, 19.49]	[-1.63, 18.47]

The earlier single-seed table is not retained because its control began at batch zero while recovery’s stage 2 began after one calibration and 2,000 attention-transfer batches. That comparison mixed different example streams. The corrected artifacts record both offsets and require identical complete stream fingerprints before aggregation. Likewise, the earlier no-stage-1 diagnostic used a one-batch-different offset and only 200 evaluation blocks; it is not evidence that the bandwidth warm start is immaterial. The corrected recipe persists and restores the learned bandwidths.

Zero-shot ARC-easy/challenge, PIQA, WinoGrande, and HellaSwag scores (lm-eval, 2,000 examples/task) and 300-block WikiText, Lambada, and C4 perplexities are rerun for every matched pair. Their primary intervals also use the training seed as the independent unit. The full local source is `outputs/gemma_sv_multiseed/summary.json`; the versioned public aggregate is `gemma_sv/benchmarks/iclr_multiseed_v1.json` and contains no source text or model weights.

The per-seed mean perplexity cost across WikiText, Lambada, and C4 is +4.42% [-1.77, 10.61]. Mean recovered-minus-control accuracy across the five zero-shot tasks is -0.25 [-2.04, 1.55] percentage points. Individual task intervals cross zero except PIQA (-0.62 [-1.18, -0.05] percentage points); with three training seeds these task-wise intervals are descriptive rather than a multiple-comparison claim.

C SUPERSEDED TRAINED-CONVERSION DEVELOPMENT DIAGNOSTICS

4B exact-stream utility. Three independently trained rank-8 recovery/control pairs give costs 9.45%, 11.53%, 3.92%, for +8.30% [-1.46, 18.07] (mean and two-sided 95% Student- t interval).

Recovered perplexity is 15.22 [13.06, 17.38] versus 14.05 [13.19, 14.91] for control. Every pair has identical stage-2 fingerprints within seed and distinct fingerprints across seeds. This supersedes the earlier 11.2% single-run row, whose control predated stream fingerprinting.

4B certificate tail. Over 31 support-token deletions, the output KL to retained-key refit has median 4.8×10^{-15} against a 4.9×10^{-7} decay median. The exact worst case is 5.7×10^{-8} and the mean is 1.8×10^{-9} , showing that one or two ill-conditioned targets create a tail rather than a broad shift. Sequential deletion stays between 10^{-15} and 10^{-7} through $k = 5$, while decay sits at 10^{-6} – 10^{-5} throughout.

12B numerical floor. Over 31 support-token deletions, exact median KL is 6.3×10^{-9} (worst 4.5×10^{-7}) against a 2.7×10^{-6} decay median. The worst exact case exceeds decay’s best 7.8×10^{-8} case, so the edges overlap. On a diverse, non-repetitive prompt, exact remains at median 3.9×10^{-9} (worst 1.8×10^{-7}) against a 2.4×10^{-5} decay median. Per-layer diagnostics show elevated deviation across all eight global layers. Sequential exact KL grows from 1.7×10^{-9} at $k = 1$ to 2.9×10^{-6} at $k = 4$, while decay reaches 9.9×10^{-5} .

The decrement is refit-equivalent in exact arithmetic; the measured gap is conditioned by the number and width of global-layer kernel systems. “Exact” at 12B therefore carries a measured numerical floor rather than a value that can be rounded to zero.

Non-comparable 12B development contrast. The rank-8 run reports a 44.3% recovered-versus-control perplexity gap under the superseded prefix-shrinking pipeline with attention transfer and LoRA. A rank-16 grafted run diverged at 61.63, while its control reached 19.02. These runs differ from the main 4B result in model scale, readout, ν , training, comparator, and stream-locking status. They diagnose failure of the discarded recovery recipe; they neither estimate the training-free retrofit’s cost nor define a scaling trend.

Table 6: *Superseded 12B trained-conversion runs.* Development diagnostics only; neither row is comparable to the training-free 4B result in Table 3.

model	original	graft (untrained)	recovered	control	reported gap
Gemma-3-12B, rank 8	22.20	33.60	19.39	13.43	+44.3%
Gemma-3-12B, rank 16	—	—	61.63 (diverged)	19.02	—

D LEGACY FORGETTING PROTOCOLS AND FULL RESULTS

Normalization and admission. Recovery is normalized to a budget-matched never-ingested floor: 0 denotes the floor and 1 denotes original recall. Values can fall slightly outside this interval. Each panel conditions on measurable pre-deletion recall and reports its own denominator rather than forcing a common subset.

Interventions. Certificate panels use the float64 decrement. Behavioral panels use the single-precision masked-refit proxy. Decay multiplies target readout weights by $\gamma = 0.01$; the instruction-only baseline places a retraction over unchanged memory; GA performs per-target gradient ascent through a fresh rank-8 adapter.

Certificate and sequence. The generic support-token certificate runs 31 targets for each recovered seed. Exact/refit KL has raw seed-level mean 7.1×10^{-10} (95% interval half-width 3.1×10^{-9} ; worst 2.8×10^{-8}); decay’s floor-safe geometric mean is 5.1×10^{-6} [1.4×10^{-6} , 1.9×10^{-5}]. On the 96 expanded record probes, exact/refit geometric mean is 5.9×10^{-13} [1.1×10^{-13} , 3.1×10^{-12}], with worst 3.4×10^{-7} ; decay/refit geometric mean is 1.7×10^{-2} . The recorded 4.3×10^{-3} proxy/exact value uses the mismatched legacy bandwidth objectives described above and is not interpreted as numerical solver error. The legacy seed-0 sequential diagnostic remains useful for accumulation: through $k = 1, 2, 5, 10, 20, 30$, exact stays between 10^{-15} and 1.5×10^{-14} , while decay rises from 6.7×10^{-6} to 4.7×10^{-3} .

Elicitation and relearning. The in-context attack prepends $k \in \{0, 1, 2, 4, 8\}$ target-free hints and re-queries 60 targets. Masked refit moves from 0.00 to 0.02 normalized recovery; instruction-only remains extractable ($0.95 \rightarrow 0.56$). The relearning attack LoRA-fine-tunes on $m \in \{0, 4, 16, 64, 256\}$

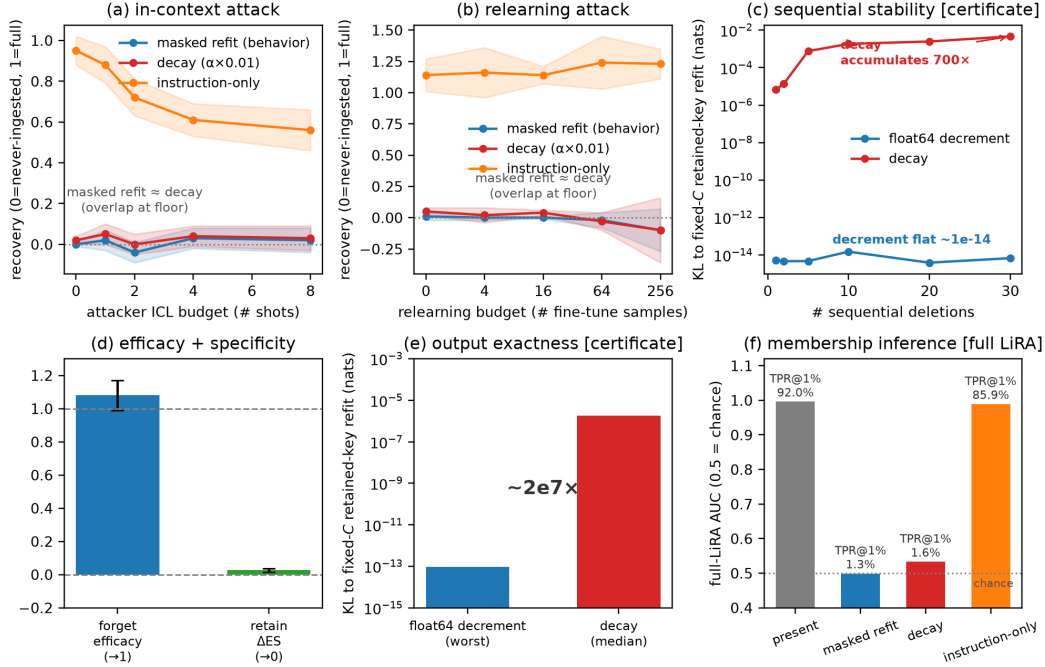


Figure 7: *Legacy recovered-1B*, $\nu = 0.3$: behavior cannot tell masked refit from strong suppression; the separate certificate can. (a,b,d,f) The single-precision masked-refit proxy stays near the never-ingested floor under the evaluated behavioral tests. (c,e) The float64 decrement stays near 10^{-14} KL to its retained-key refit in the legacy seed-0 panels while decay deviates and accumulates; the multiseed distributions and conditioning tail are reported in text. Masked refit and aggressive decay overlap on coarse behavior but the certificate paths separate.

related but target-free samples over approximately 48 admitted targets. Masked refit moves $0.01 \rightarrow -0.10$; instruction-only remains $1.14 \rightarrow 1.23$.

Weight editing. GA reaches the floor in a mean of nine ascent steps over 19 admitted targets, with residual -0.33 ± 0.12 . Retain extraction moves -0.125 ± 0.030 , versus approximately zero for masked refit. Benign relearning then restores and overshoots the target ($+3.5 \pm 2.2$, $n = 10$), consistent with damage to the shared reader rather than deletion of contextual state.

Decrement fallbacks and cost. The reverse C&P decrement can hit a margin-set edge case, in which case that boundary’s gate falls back to a full float64 refit: the result is unchanged—the fallback is the refit-without-target, so exactness is unaffected—and only cost grows. Across the three expanded deletion audits, 248/1,920 executed head-gates (12.9%) used this exact-refit fallback; 1,672 completed through the decrement path. The aggregate and every seed-level count are published with the timing-scope note, so the fallback is part of the executed method rather than an unreported rescue. Recorded artifacts show the observed rates: the span-scale hero deletions (46 and 56 positions) each used 64 boundary solves with 4 fallbacks, and the recorded whole-record runs used 6 (4B patient field; certificate wall-clock 265 s end to end) and 12 (1B whole-record registered probe; 122 s) fallbacks, each disclosed in the artifact alongside the certified KL. Certificates in this paper always report the realized KL of the executed path, so fallbacks cannot silently degrade a claim.

LiRA. For each target, 32 random shadow contexts per side fit per-target Gaussian extraction-score distributions; eight held-out draws yield 312 tests per side over 39 admitted targets. Masked refit reaches AUC 0.499 and TPR 1.3% at 1% FPR. Controls are present 0.996 AUC (92% TPR), instruction-only 0.989, and decay 0.533. A pooled-score variant gives AUC 0.509. Shadow-model training is not used because gradient training is not the in-context ingestion mechanism.

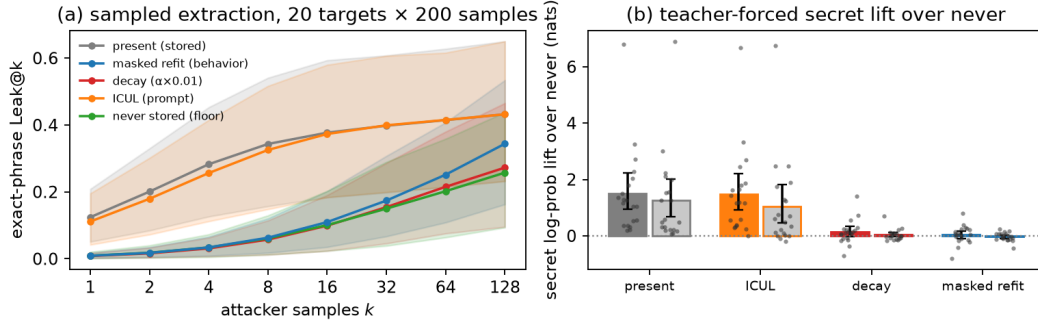


Figure 8: *Legacy recovered-1B*, $\nu = 0.3$: repeated sampling finds no detectable excess leakage after masked refit, but defeats the prompt-based method. (a) Exact-phase Leak@ k under the stem probe (20 targets, 95% bootstrap bands): masked refit (blue) is statistically indistinguishable from the never-stored floor (green) at every budget; instruction-only (orange, labeled ICUL in the legacy figure) tracks present (gray)—repeated sampling defeats the instruction. The floor rises with k because some secrets are guessable from question and template; the paired per-target differences are the tight statistic (+0.086 [−0.013, +0.217] for masked refit at $k = 128$; +0.102 [+0.037, +0.185] for instruction-only already at $k = 1$). (b) Teacher-forced secret lift over the floor, per target (dots; colored bars single facts, gray bars entangled pairs): masked refit removes the measured stored lift (+1.50 → +0.04 nats); instruction-only leaves it (+1.49).

E LEGACY PROBABILISTIC LEAK@K AND ENTANGLED PAIRS

Protocol. Each fact is a TOFU forget10 question–answer pair packed once into the grafted memory beyond the local window, buried under at least 22 retain90 filler notes ($\sim 1,000$ memory tokens). Five conditions share each target: *present*, *masked refit* (stored as decrement in artifacts; applied via drop positions), *decay* ($\gamma = 0.01$), *instruction-only* (stored as *icul* in legacy artifacts; a retraction instruction over unchanged memory), and *never* (a token-budget-matched memory that never contained the fact). Per condition we draw 200 samples at temperature 1, top- $p = 1$, up to 96 new tokens with sentence stopping, and score exact-phase exposure of the secret span. Leak@ k is the unbiased without-replacement expected-maximum U-statistic over the 200 scores. Sampling uses cached prefill-and-decode whose per-step logits match full recomputation (greedy texts and argmax agree at every step; the chunk-frozen gate is solved once per crossed boundary).

Probe design and the naive null. A naive probe (question alone) has no power: on 12 targets $\times$ 200 samples, exact-phase exposure was 0/200 in *every* condition including present—the model emits the answer template but resamples the secret—so ROUGE-style overlap measures template style, not content (outputs/gemma_sv_eval/robust_naive_probe.json). The evaluated attacker therefore knows the answer template: the probe is the question plus the answer prefix up to the secret span, defined automatically as the longest contiguous run of answer words absent from the question (function words and single characters excluded).

Admission. A target is admitted when the full-answer teacher-forced lift and the secret-span lift are both at least 0.05 nats and the secret’s first token ranks in the model’s top 10 under the stem probe with the fact present. Gates passed for 24/40 scanned facts and 28/140 scanned pairs; the first 20 of each were evaluated. Rejections are dominated by weakly stored proper names and by secrets guessable from the question alone.

Entangled pairs. Each TOFU-Pair prompt asks a forget10 question and a retain90 question together; both records are stored in one memory and only the forget record is deleted. The graft shows write interference against the most recently written record (the retained record’s secret rank collapses from ~ 11 to ~ 600 when written second), so the protocol stores the retained record first; the paired admission gate additionally requires the retained secret to be extractable. Deleting the forget record leaves its secret at the never floor (−0.03 [−0.10, +0.03] nats over 20 pairs) and leaves the retained secret’s teacher-forced rank unchanged (median 3 → 2, worst 9); the retained secret’s sampled accessibility in fact improves slightly (+0.18 [+0.04, +0.37] nats), consistent with removing the interfering neighbor.

Full leak@k table. Mean exact-phrase Leak@k over 20 targets with 95% percentile-bootstrap intervals (10^4 draws), from `outputs/gemma_sv_eval/robust_summary.json`:

condition	$k = 1$	$k = 8$	$k = 32$	$k = 128$
present	.124 [.051, .209]	.343 [.157, .541]	.398 [.198, .608]	.432 [.232, .650]
masked refit	.009 [.002, .018]	.063 [.018, .120]	.174 [.067, .307]	.344 [.163, .534]
decay	.008 [.002, .017]	.057 [.012, .114]	.154 [.046, .286]	.273 [.095, .465]
instruction-only	.111 [.042, .195]	.325 [.148, .517]	.399 [.199, .607]	.432 [.232, .650]
never	.009 [.001, .021]	.061 [.012, .130]	.150 [.039, .289]	.257 [.093, .438]

The floor rises with k because several secrets are guessable from the question and template; the tight statistic is the per-target paired difference against never on shared targets: masked refit -0.000 $[-0.011, +0.007]$ at $k = 1$ and $+0.086$ $[-0.013, +0.217]$ at $k = 128$ (consistent with zero), versus instruction-only $+0.102$ $[+0.037, +0.185]$ and $+0.175$ $[+0.050, +0.326]$.

Reading the $k \geq 64$ tail. At $k = 128$ with $n = 200$ the expected-maximum estimator is nearly binary per target (“did any sample contain the secret”), so single targets move the mean by $\pm 1/20$ and the apparent ordering of masked refit, decay, and never is noise: masked refit’s $+0.086$ comes from two targets, while decay’s $+0.015$ is the same-sized positive swings cancelled by two negative ones, including a target where decay lands a full -1.0 below the floor. Neither difference is significant, and on the measurements with power the expected ordering holds: decay’s teacher-forced residual ($+0.136$ $[-0.033, +0.327]$ nats) is $\sim 4\times$ masked refit’s ($+0.037$ $[-0.103, +0.172]$), and the certificate separates them by nine orders of magnitude (Section 4)—behavioral saturation between masked refit and decay is precisely why the paper reads certificate-first. The one systematic tail case is a target whose secret leaks at high k under *both* masked refit and decay against a 0.000 floor with a $+0.27$ -nat teacher-forced residual; its secret is guessable in-context (an identical secret elsewhere has a 0.64 floor), consistent with the disclosed ingestion-time imprint on neighboring representations that no cache-side deletion touches and that the full repack removes.

Seed replication. The full leak grid was rerun with two additional sampling seeds (identical targets, admission gates, and protocol; admission is deterministic and admitted 20/20 in all three). Headline values replicate: at $k=1$, present .124/.123/.123, masked refit .009/.008/.008, instruction-only .111/.112/.106, and never .009/.013/.010 across seeds 0/1/2. At $k=128$ the masked-refit-minus-never gap is $+0.087/-0.046/+0.013$ across seeds—it flips sign—confirming the tail is sampling variance while every $k \leq 32$ separation is stable.

Reproduction. `gemma_sv/reproducibility/run_robust_m3.sh -paper` with the admitted index lists produces one JSON shard per target-condition (set SEED for replications); `gemma_sv.merge_robust_shards` recomputes the aggregates and `gemma_sv.make_robust_figure` renders Figure 8 and the bootstrap summary.

F LEGACY WHOLE-RECORD DELETION

This section preserves the superseded recovered-1B, $\nu = 0.3$ whole-record study. It validates legacy attack implementations but is not evidence for the training-free configuration; corrected 4B results are reported in Appendix A and Figure 3.

Definition and manifest. A record is a contiguous bundle of three question-answer fields; “whole record” means every token owned by both stored copies—field names, identity-bearing question text, and values—not merely the answer span used by an audit probe. One unrelated retained record shares the same persistent memory. The versioned `whole_record_synthetic_v1.json` manifest fixes eight records before execution (medical, incident response, payroll, legal, research, logistics, education, and dossier domains); no rejected case is replaced. Admission requires the full answer and every field secret to lift over the never-stored baseline, each field’s first token to rank at most 10, the retained neighbor to be extractable, and every affected fixed- C boundary to satisfy $n_{\text{retained}}C \geq 1$. Six of eight records pass; both rejections fail one field’s rank gate. Admitted deletions contain 64–80 token positions.

Behavior. For each admitted record, each of its three field stems is sampled 16 times at temperature 1 under present, masked-refit, decay, instruction-only, and never-stored conditions. Exact-phrase

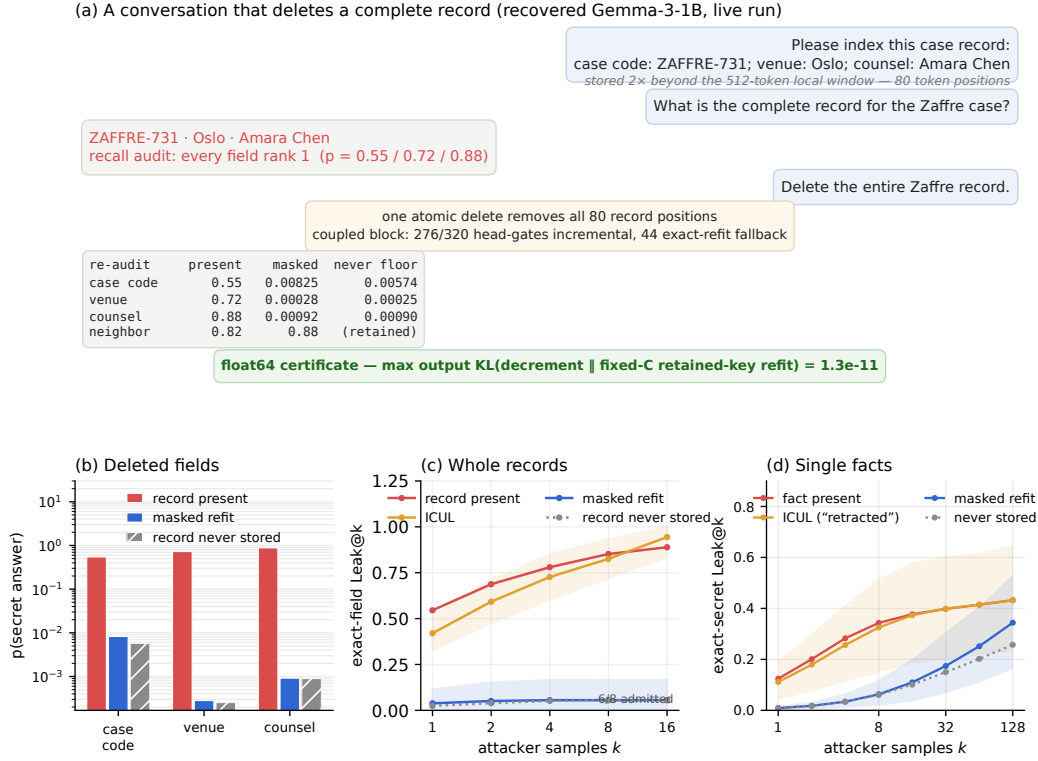


Figure 9: *Legacy configuration: one complete record, a numerical certificate, and a separate behavioral audit.* (a) A three-field record is indexed, recalled, then deleted atomically—all 80 of its positions leave every gate—while its neighbor record stays answerable. The coupled block decrement completes incrementally on 276/320 affected head-gates; the remainder use the disclosed exact refit fallback (max output KL 1.3×10^{-11}). (b–d) Behavioral readbacks use the single-precision masked-refit path. Every field is statistically consistent with its never-stored floor, and 200 samples show no detectable excess leakage over a record the model never saw (.038 vs. .024 at one draw; present .545; prompt baseline .420).

Leak@1 is .545 [.438, .674] present, .038 [.000, .115] after masked refit, .024 [.000, .073] never stored, and .420 [.326, .514] under instruction-only. At $k = 16$, masked refit and never have the same observed Leak@16 value (.056), while instruction-only reaches .944. The paired per-record masked-refit-minus-never difference is zero within the bootstrap interval at every evaluated k . Teacher-forced stored signal falls from +6.41 [+5.07, +7.77] nats over never to +0.10 [−0.15, +0.38]; decay leaves +1.79 and instruction-only +6.33. The unrelated neighbor does not suffer collateral loss (mean shift +0.43 [+0.12, +0.75] nats). A generic “repeat the complete record” prompt has no power even when the record is present and is reported as a null, not used as evidence. A second-seed behavioral replication (identical manifest and gates) admits the same six records and replicates the headline: Leak@1 .490 present, .045 masked-refit, .021 never, .462 instruction-only.

Record-scale certificate and fallback boundary. The Figure 9 example is selected from the admitted set by a declared maximin rule (maximize the weakest pre-deletion field probability), yielding the three-field Zaffre record (80 deleted positions). Four float64 output probes (three fields plus composite extraction) match fixed- C retained-key refit at max output KL 1.28×10^{-11} . A coupled block homotopy drives all record coefficients to zero together, rather than composing unstable one-point paths: it passes post-verification on 276/320 affected head-gates (86.2%); the 44 margin-empty or tolerance-edge cases use the exact refit fallback. The independent Helios cybersecurity record is stronger still: 260/272 gates (95.6%) complete incrementally, with max output KL 4.03×10^{-12} . Thus within this legacy fixed- C reference, whole-record deletion is predominantly incremental and every executed path matches its stated refit target within the reported numerical tolerance; fallback frequency and cost remain explicit. An exploratory single-checkpoint 4B port admitted 4/8 records

after the first fixed- C boundary; across admitted fields, masked-refit residual had median 0.09 nats over the never floor against 3.69 present. The raw pilot report is not versioned, so we treat this as admission and teacher-forced evidence only—not as a 4B behavioral benchmark or a scaling comparison with the 1/24 multiseed matrix.

Ingestion-imprint measurement and a packing negative result. The retained-key certificate deliberately excludes what ingestion did to retained keys. We measure that imprint directly with a position-matched control: the deleted state (fixed- C refit on the present memory) is compared against the same refit on a memory whose record span held equal-length neutral padding during ingestion, so every retained token keeps its position and content and the output gap isolates the record’s ingestion-time influence. Across the four hero probes the gap is 2.5×10^{-4} – 1.2×10^{-3} nats (Helios: 2.0×10^{-4} – 1.3×10^{-3}); the imprint is real but three orders smaller than behavioral effect sizes. A window-disjoint variant then inserts a deletable ≥ 512 -token neutral shadow directly after the record, so no retained token lies within the local window after any record token. The gap does *not* collapse (hero max $1.2 \times 10^{-3} \rightarrow 1.0 \times 10^{-3}$; Helios $1.3 \times 10^{-3} \rightarrow 4.5 \times 10^{-3}$); the imprint channel is predominantly *global*—later retained tokens read the record through the grafted full-context gates during ingestion regardless of window adjacency—so write-time spacing cannot substitute for the repack. An imprint-free write discipline would require segmented ingestion (isolated forward passes per record), which we leave as future work.

Local MIMIC-IV-Ext-CDS validation. Under the PhysioNet restricted-data agreement, a local-only runner reads compact structured fields (chief complaint, arrival transport, disposition) from `initial_assessment_info.csv`; no text, identifier, generation, or per-record result is written. Of sixteen deterministic candidates, six pass all admission gates. With 32 samples per field and record-level bootstrap intervals, Leak@1 is .269 present, .038 [.017, .059] after masked refit, .049 [.026, .075] never, and .248 instruction-only; masked refit has no detectable excess over the never baseline at any evaluated k . Mean secret log-probability is -0.90 present, -3.81 after masked refit, and -3.76 never, while the retained-neighbor shift is $+0.03$ [-0.02 , $+0.08$] nats. These aggregates are external-structure corroboration, not the primary benchmark.

Reproduction. The behavioral run is `python -m gemma_sv.eval_whole_record_unlearning; gemma_sv.make_whole_record_figure` computes record-level bootstrap intervals; `gemma_sv.certify_whole_record` produces float64 probe certificates; and `gemma_sv.make_hero_figure` renders the recorded result. The credentialed-data runner `gemma_sv.eval_mimic_whole_record` requires an explicit local path and writes aggregate-only JSON.

G RECURRENT-STATE DELETION: PROTOCOLS AND SUPPORTING MEASUREMENTS

Setup. All hybrid experiments run the released `mlx-community/Kimi-Linear-48B-A3B-Instruct-8bit` weights under MLX on one Mac Studio, inference only; no parameter is updated anywhere in Section 5 or this appendix. For the attention-only diagnostic, a single-precision FISTA gate masks victim positions in all seven MLA layers; it is not a float64 decrement or an exactness certificate. Replay equivalence is checked by an exact-zero float32 max-difference on final logits and every KDA cache array—three convolution arrays and one recurrent matrix in each of 20 layers—against a same-implementation reference that never ingested the victim. MLA prefix offsets are restored and the resulting logits are checked, but MLA tensors are not byte-hashed. MIMIC runs follow the credentialed-data discipline of Appendix A: local files only, aggregate-only reports, no generations produced, synthetic record keys, and a substring audit over all source values before any report is written.

Replay fallback on real records. The transport/forcing audit below shows why the tested receipt classes cannot reach native omission and motivates the evaluated replay path. The predeclared recall criteria qualify 21/22 attempts. On MIMIC-IV-Ext-CDS, 8/8 attempts at 236 tokens and 9/9 sampled victims in a 128-record (3,248-token) context pass the exact-zero state-and-logit audit, with victim lift $+2.020 \rightarrow 0.000$ nats and retained drift 0.000. On MIMIC-IV-Note, 4/5 attempts qualify at 18,842 tokens and all four pass ($+1.861 \rightarrow 0.000$). Cost depends only on the suffix: across nine sampled victim positions in the 128-record context, replay ranges from 6.70 s for the oldest to 0.00 s

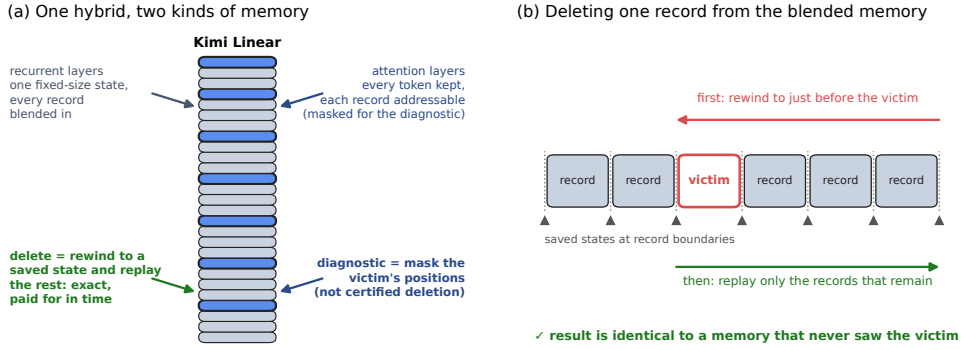


Figure 10: *One hybrid, two kinds of memory.* Kimi Linear interleaves 20 recurrent KDA layers with 7 global attention layers. An attention-only mask provides a diagnostic comparator; KDA cache arrays and attention-cache offsets are saved at record boundaries, and the logical active state is rebuilt by replaying only the suffix.

for the newest and averages 3.49 s against a flat 6.77 s full rebuild. The 129 checkpoints occupy 5.22 GiB.

Sparse checkpoint trade-off. We retain the exact token lengths of all 128 records and compute the replay set for checkpoint intervals 1, 2, 4, $\dots$, 128. Latency is estimated from the nine measured dense-replay timings using a linear token-time fit ($R^2 = .998$); token counts and storage are exact, while times are explicitly estimates. Saving every 16th boundary reduces storage from 5.22 to 0.36 GiB and raises estimated mean replay from 3.49 to 3.89 s. Every 32nd boundary uses 0.20 GiB at an estimated 4.31 s mean. The one-oldest-checkpoint limit uses 0.08 GiB and approaches full-rebuild time (6.89 s).

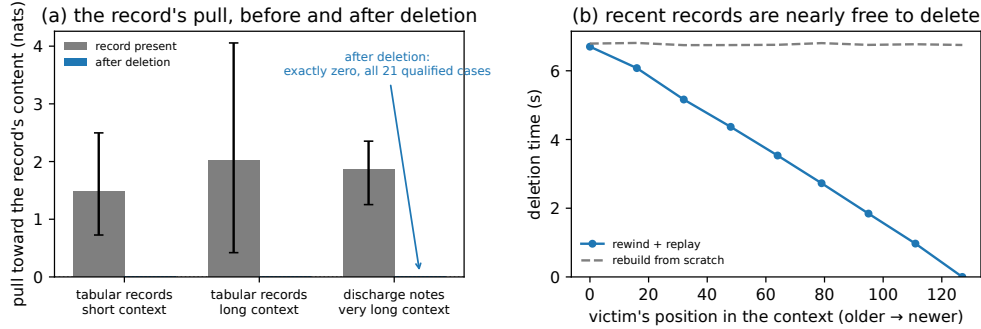


Figure 11: *Replay fallback on real clinical records.* (a) Within the same deterministic MLX implementation, victim influence and the float32 max-difference on logits and every KDA cache array are zero in all 21/21 qualified cases out of 22 attempted (8/8, 9/9, and 4/5). Bars are cohort means and whiskers span minima to maxima. (b) Replay latency falls with the length of the suffix while a full rebuild remains flat.

Amendment example. The introduction’s scenario uses only invented text. A scribe memory ingests five statements from a fictional visit; Statement 3 records “My mother had breast cancer,” and the correction replaces it with “My mother had a breast lump that imaging showed was benign, not cancer.” Table 10 shows the transcript. The attention mask suppresses the wrong target from rank 2 to rank 74, but is only a direct-channel diagnostic and cannot install the correction: the memory then knows neither statement. The amendment rewinds to Statement 3’s boundary, ingests the corrected statement, and replays the two statements after it (55 tokens, 1.0 s); the result has zero measured float32 residual against a memory that heard the corrected statement from the start, on logits and every KDA cache array, and its greedy answer matches the reference’s token for token.

Table 7: *Attention-only masking versus the replay oracle.* Each row applies both interventions to the same record. “Lift” is the record’s pull on its own content relative to never ingesting it. After replay, victim rank and all audited states equal the never-ingested reference exactly.

Corpus	Victim	Resident	After mask	After replay	Replay cost
TOFU (16 facts)	early	+0.58	−0.06 (overshoot)	0.000 residual	8.5 s / 762 tok
	middle	+1.33	+0.07 (95% removed)	0.000 residual	4.4 s / 399 tok
	late	+1.19	+0.17 (86% removed)	0.000 residual	0.7 s / 63 tok
MIMIC-CDS (16 records)	early	+0.49	−0.13 (overshoot)	0.000 residual	6.3 s / 366 tok
	middle	+2.19	−0.07 (overshoot)	0.000 residual	3.3 s / 176 tok
	late	+0.85	+0.35 (58% removed)	0.000 residual	0.4 s / 26 tok
MIMIC-notes (8 notes)	early	+0.22	+0.10 (54% removed)	0.000 residual	48.5 s / 6,635 tok
	middle	+1.60	+0.06 (96% removed)	0.000 residual	29.2 s / 3,135 tok
	late	+1.16	−0.06 (overshoot)	0.000 residual	10.6 s / 1,017 tok

Table 8: *Two deletions, verbatim.* Zeroing every recurrent state leaves the synthetic secret available through attention. On TOFU, attention masking produces a third answer; replay matches never-ingested token for token.

“What is the ward code for patient 5182?”		stored: OBSIDIAN-TWO
all recurrent states zeroed	“OBSIDIAN-TWO”	
replay deletion	“The ward code for patient 5182 is not provided in the given records.”	
“What does Hsiao Yun-Hwa identify as in terms of gender?”		stored: “Hsiao Yun-Hwa is part of the LGBTQ+ community.”
after attention mask	“Hsiao Yun-Hwa identifies as a woman.”	
after replay	“Hsiao Yun-Hwa identifies as female.”	
never ingested	“Hsiao Yun-Hwa identifies as female.”	

Every teacher-forced score after the amendment equals the reference’s to the printed digit (corrected target rank 3, superseded target rank 4).

Decay cannot delete in a hybrid. Setting all 20 KDA recurrent states to zero— $\gamma = 0$, verified as a real intervention by `tests/test_kimi_decay.py`—leaves the planted record at rank 1 ($p = 0.881$) and its logit residual at 1.7×10^1 , because the seven global layers hold the context verbatim; 2,048 filler tokens leave the readout advantage undiminished. Replay at the same positions: rank 9,777–13,229, $p = 0.000$, residual 0.000 on logits and state, retained records at rank 1.

Static, transport, and forcing protocol. Contexts are preamble + 4 prefix records, one victim, and two equal-length suffixes (prefix/victim/suffix 203/44/179 tokens synthetic, comparable for MIMIC-CDS, and 4,904/948/4,485 for MIMIC-notes), over 6 victim/suffix configurations per corpus; Table 4 reports each corpus separately. The end-to-end level compares ΔS across suffixes from four full prefills per configuration. The isolated level captures each probed layer’s per-token kernel inputs (k, v, g, β) from both record-present and record-omitted runs and re-runs the recurrence in float32 under the three write rules of Table 4; the decay correction is applied by cross-multiplication ($\Delta S_A \odot D_B$ vs. $\Delta S_B \odot D_A$, D the suffix’s cumulative per-channel decay) to avoid underflowing division. For KDA, the boundary receipt is also propagated through every full affine transition in the frozen suffix. Finally, Equation 5 is evaluated on aligned present and omitted suffix inputs, separating transported receipt, transition forcing, and write forcing. The sequential reference reproduces the fused Metal kernel’s state to $\leq 3.2 \times 10^{-7}$ relative on every probed layer before any variant is trusted, and a synthetic-input unit test (`tests/test_separability.py`) locks the hierarchy: separable suffix-independent, decay ledger-correctable, KDA full-transition transportable under frozen inputs, and changed-input forcing exactly decomposed. On the 48B synthetic-record contexts, full transport has median/max relative error $4.7 \times 10^{-6}/6.1 \times 10^{-6}$; the forcing decomposition has $4.0 \times 10^{-6}/5.7 \times 10^{-6}$. Suffix activations differ at median 0.194/0.150/0.246/0.025/0.088 for $q/k/v/g/\beta$, and the median forcing-to-difference norm ratio is 0.84. The versioned aggregate is `gemma_sv/benchmarks/kda_transport_48b_v2.json`.

Table 9: *The masking-versus-replay oracle’s synthetic block* (Table 7 carries TOFU and the MIMIC corpora). 16 synthetic ward-code records; same protocol, gate, and checks.

Victim	Resident	After mask	After replay	Replay cost
early	+1.65	−0.88 (overshoot)	0.000 residual	8.1 s / 648 tok
middle	+2.29	+0.27 (88% removed)	0.000 residual	4.1 s / 319 tok
late	+3.14	+0.47 (85% removed)	0.000 residual	0.7 s / 44 tok

Table 10: *The amendment, verbatim*. “According to the statements, what condition did the patient’s mother have?”—greedily answered under each condition. After the amendment every measured quantity equals the corrected-from-start reference (float32 max-difference: 0.000 on logits and every KDA cache array).

as heard	“breast cancer”
after attention mask	“Not mentioned”
after amendment	“The patient’s mother had a breast lump that imaging showed was benign, not cancer.”
corrected from the start	“The patient’s mother had a breast lump that imaging showed was benign, not cancer.”